No.	Solution
1(a)(i)	The force in rope B must not have a horizontal component. Therefore force Y must act vertically upwards at an angle of 90^0 and equal to magnitude of weight of S at 60 N.
1(a)(ii)	Horizontal component of x = 200cos30 Vertical component of x = 200sin30 Resolve horizontally, $Y_{horizontal} = 200\cos 30 = 173.2\text{N}$ $\text{Y}_{vertical} + 200\sin 30 = 60\text{N}$ $Y_{vertical} = -\ 40\text{N}$ *magnitude* of Y = $\sqrt{40^2 + 173.2^2} = 178\text{ N}$ angle of θ= -13.0^0 or 13.0^0 below horizontal
1(b)	Force x has a horizontal component pointing towards the right. In order to achieve equilibrium, this horizontal force has to be balanced by a leftward force by rope B. If rope B is parallel to S, there will not be a horizontal force available and thus equilibrium will not be achieved.
2(a)	P type semiconductor is doped with Group 3 impurities with 1 less valence